Project Report: F1 Racing Intelligence & Telemetry Analytics Platform

Team ID: PNT2022TMIDXXXXXX **Date:** 24 September 2026

Project Name: F1 Racing Intelligence & Telemetry Analytics Platform

1. Introduction

1.1 Project Overview

The F1 Racing Intelligence & Telemetry Analytics Platform is an enterprise-grade cloud analytics solution built to ingest, process, and visualize massive streams of Formula 1 telemetry, race timing, and historical race data. Leveraging Snowflake data cloud and Streamlit/Tableau visualization interfaces, this platform transforms complex multi-gigabyte race logs into actionable performance intelligence for race strategists and engineering teams.

1.2 Purpose

The primary objective is to provide real-time race telemetry analysis, tire degradation forecasting, and pit-stop optimization. By centralizing disparate timing intervals and weather variables into a unified data warehouse, the system eliminates latency in tactical decision-making during high-stakes Grand Prix events.

2. Ideation Phase

2.1 Problem Statement

Modern Formula 1 teams generate gigabytes of telemetry data per second per car. Manually correlating live weather conditions, tire wear profiles, and lap time deltas leads to split-second strategic errors, sub-optimal pit windows, and compromised race finishes.

2.2 Empathy Map Canvas

- **Says:** "We need immediate visibility into tire graining and degradation rates before competitors pit."
- **Thinks:** "If we miss the undercut window by even one lap, we lose podium contention."
- **Does:** Continuously monitors multi-channel telemetry dashboards and radio chatter under high-pressure scenarios.
- **Feels:** Anxious about unpredictable weather shifts and unexpected mechanical safety cars.

2.3 Brainstorming

Core ideas generated during the ideation phase included automated anomaly detection for power unit telemetry, predictive tire degradation modeling via historical asphalt friction coefficients, and automated pit-stop efficiency scoring.

3. Requirement Analysis

3.1 Customer Journey Map

The operational workflow maps race engineers from initial session briefing through live qualifying and race execution, culminating in post-race telemetry debriefs and setup adjustments.

3.2 Solution Requirement

- **Functional:** Real-time ingestion of lap times, sector splits, GPS coordinates, and weather metrics.
- **Non-Functional:** Sub-second dashboard query response times, 99.9% pipeline availability, and secure role-based access.

3.3 Data Flow Diagram

Raw JSON telemetry feeds and CSV timing dumps are staged in Snowflake internal stages, transformed via SQL stored procedures into curated dimensional models, and served directly to interactive visualization frontends.

3.4 Technology Stack

Layer	Technology / Tool	Purpose
Data Warehouse	Snowflake Data Cloud	Scalable storage and high-concurrency SQL compute.
ETL / Processing	Python (Snowpark / Pandas)	Data cleansing, telemetry aggregation, and feature engineering.
Visualization	Streamlit & Tableau	Interactive dashboards and real-time telemetry plotting.

4. Project Design

4.1 Problem Solution Fit

By decoupling compute and storage in Snowflake while providing low-latency analytical views, the platform bridges the gap between raw multi-gigabyte sensor streams and instantaneous pit-wall strategy execution.

4.2 Proposed Solution

A modular data pipeline architecture featuring dedicated raw staging layers, curated analytics models, and an interactive multi-page dashboard application designed for race strategists.

4.3 Solution Architecture

The architecture consists of three core tiers: Data Ingestion (REST APIs & CSV bulk loads), Cloud Storage & Transformation (Snowflake Snowflake SQL & Snowpark), and Presentation (Streamlit UI components).

5. Project Planning & Scheduling

5.1 Project Planning

The project was executed over a 4-week sprint cycle: Week 1 (Requirement Gathering & Snowflake Schema Design), Week 2 (ETL Pipeline Development & Data Cleansing), Week 3 (Dashboard Visualization & Query Optimization), and Week 4 (Performance Testing & Deployment).

6. Functional and Performance Testing

6.1 Performance Testing

Stress tests were conducted simulating 50 concurrent engineering users querying multi-season telemetry tables. Snowflake virtual warehouse auto-scaling maintained sub-second query execution across all analytical views.

7. Results

7.1 Output Screenshots

The dashboard successfully renders live lap time distributions, top-speed comparisons under wet vs. dry conditions, and constructor championship points progression across seasons.

8. Advantages & Disadvantages

Advantages

- Centralized single source of truth for all historical and live race telemetry.
- High concurrency support enabling multiple engineers to analyze data simultaneously without latency.

Disadvantages

- Reliance on cloud internet connectivity during live track sessions.
- Initial cloud compute credit management required careful warehouse sizing.

9. Conclusion

The F1 Racing Intelligence & Telemetry Analytics Platform successfully demonstrates how modern cloud data warehouses like Snowflake combined with robust BI tools can revolutionize race strategy, reduce human calculation error, and provide a competitive edge on race day.

10. Future Scope

Future enhancements will integrate machine learning models for predictive tire puncture detection, automated audio transcription of team radio comms, and integration with live FIA timing feeds.

11. Appendix

- **Source Code:** GitHub Repository
- **Dataset Link:** Snowflake Data Marketplace / F1 Datasets
- **Project Demo:** Live Streamlit App